TaskFlow

Plan • Organize • Achieve

Functional Requirements

Personal & Collaborative Project Management Platform

PROGRAM	**DEPI Full Stack .NET Project**
DOCUMENT	**Functional Requirements**
VERSION	**1.0**
PREPARED	**15 October 2026**
STATUS	**Draft for team review and lecturer feedback**
AI	**Out of scope**

1. Purpose

This document defines the functions TaskFlow is expected to provide to its users. It translates the project proposal and the agreed product scope into identifiable, testable requirements. Each requirement has an ID, a priority, and a concise verification condition to support implementation, testing, and traceability.

The source project specification focuses on workspaces, teams, projects, Kanban boards, tasks, real-time collaboration, Gantt charts, time tracking, analytics, notifications, activity logs, automation, and administration. The team has also agreed to include a personal-first experience and selected collaboration extensions described below.

2. Product Scope

TaskFlow is a project and task management web application for **individual users, freelancers, small teams, and companies**. A user must be able to organize personal work without first creating a team. Users can later create or join a workspace when they need to collaborate.

The system will provide account and access management, personal work management, workspace and team administration, project planning, task execution, collaboration, time tracking, reporting, notifications, rule-based automation, and administrative tools.

2.1 Scope boundaries

- **No AI features are in scope.** TaskFlow's automation and health alerts will use explicit user-configured rules and measurable conditions, not AI-generated predictions or recommendations.
- Public marketing pages such as Landing, Features/How It Works, Pricing, and optional Contact/FAQ pages are being handled separately and are **not included in the internal module allocation in this document**.
- This document defines desired product behavior. It does not claim that any feature has already been implemented, tested, deployed, or approved by the lecturer.
- Azure deployment, monitoring, database configuration, and CI/CD are technical delivery requirements. The user-facing functions they support are described here; infrastructure configuration itself is not a user functional requirement.

3. Actors and Permissions

Actor	Description	Typical permissions
Individual User	Uses TaskFlow for personal tasks and projects	Manage own profile, personal workspace, tasks, views, and time records
Freelancer	Manages independent work and may share selected project information with clients	Personal features plus client/project sharing where enabled
Workspace Owner	Owns a workspace and controls its main configuration	Manage workspace, members, roles, settings, projects, and administrative actions
Workspace Admin	Helps administer a workspace	Manage permitted members, projects, and settings according to granted permissions

Actor	Description	Typical permissions
Team Member	Collaborates on assigned or shared projects	Work with permitted projects, tasks, comments, files, and time entries
Viewer	Has read-only access to specified workspace/project data	View authorized information; no write actions unless explicitly granted
Project Manager / Team Lead	Plans work and follows delivery and workload	Manage permitted projects, task assignments, project progress, reports, and rules
Client	External stakeholder invited to a designated client-facing area	View only explicitly shared project information and respond to permitted reviews or requests
Platform Administrator	Operates the TaskFlow platform	Manage platform-level users, settings, usage, and administrative records

Authorization rule: The actions available to a user depend on their role, project/workspace membership, and the specific permissions granted. A role name alone must not expose records the user is not authorized to access.

4. Requirement Priorities

- **Must (P0):** Essential for a credible first release and core project demonstration.
- **Should (P1):** Important for the intended product, but may be scheduled after the core workflow if time is constrained.
- **Could (P2):** Valuable enhancement that may be implemented if time and resources allow.

These priorities are the team's proposed planning priorities, not an official DEPI grading rubric. Final priorities should be confirmed by the team and lecturer.

5. Functional Requirements

For every requirement, “Verification” describes a minimum observable result that can be checked during testing. Exact UI details may be finalized in the System Analysis & Design phase.

5.1 Account, Authentication, and Personal Settings

ID	Priority	Functional requirement	Verification / acceptance condition
FR-AUTH-01	Must	The system shall allow a visitor to register an account using required identity and sign-in information.	Valid details create an account; invalid or duplicate details show a useful validation message.
FR-AUTH-02	Must	The system shall allow an existing user to sign in and sign out securely.	Valid credentials start a session; invalid credentials do not; signing out ends the authenticated session.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-AUTH-03	Should	The system shall provide an account recovery flow for users who cannot sign in.	A valid recovery request follows the configured reset process without revealing whether an unrelated account exists.
FR-AUTH-04	Should	The system shall support email verification where enabled for the deployment.	An unverified account is shown its verification state and can complete the configured verification flow.
FR-AUTH-05	Must	The system shall enforce authorization on protected pages and actions.	A user without required permissions is denied access to both the UI action and the protected server operation.
FR-AUTH-06	Must	The system shall allow users to view and update permitted profile information.	Valid changes are saved and displayed on the user's profile.
FR-AUTH-07	Should	The system shall provide account security and preference settings, including password change and notification preferences.	A valid change is saved; invalid input is rejected with a clear message.
FR-AUTH-08	Should	The system shall allow a user to select a preferred visual mode, including light/dark mode.	The selected mode is applied consistently across supported application pages.

5.2 Personal Workspace and Productivity

ID	Priority	Functional requirement	Verification / acceptance condition
FR-PER-01	Must	The system shall provide a personal workspace for an individual user without requiring team creation.	A newly registered user can reach a personal work area before joining or creating a team.
FR-PER-02	Must	The system shall provide a personal dashboard showing relevant tasks, due dates, and progress.	The dashboard displays only records the signed-in user is allowed to see and reflects saved task changes.
FR-PER-03	Must	The system shall provide a “My Day” or equivalent view for work due today and work selected by the user.	The user can identify today's relevant tasks and open each task from the view.
FR-PER-04	Must	The system shall provide a “My Work” view across the user's authorized personal and collaborative work.	Tasks assigned to the user appear with status, priority, and due-date information when available.
FR-PER-05	Should	The system shall provide Quick Capture for quickly recording a task or idea with minimal required input.	A submitted item is saved and can later be completed with additional details; task business rules remain centralized in the task module.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-PER-06	Should	The system shall allow users to customize supported dashboard widgets or the order in which information is shown.	Supported preferences are saved and restored on a later visit.

5.3 Workspaces, Teams, Membership, and Invitations

ID	Priority	Functional requirement	Verification / acceptance condition
FR-WS-01	Must	The system shall allow an authorized user to create and configure a workspace.	A workspace is created with a name and owner, and appears only to authorized users.
FR-WS-02	Must	The system shall allow authorized users to create and manage teams within the supported workspace model.	An authorized user can create, rename, and view permitted team records.
FR-WS-03	Must	The system shall allow an authorized workspace administrator to invite people to join a workspace or team.	A valid invitation is created with the intended role/permissions and can be accepted or declined according to its state.
FR-WS-04	Should	The system shall support invitation status tracking and cancellation by an authorized inviter or administrator.	Pending, accepted, expired, or cancelled invitations show the correct state; unauthorized users cannot manage them.
FR-WS-05	Must	The system shall support the roles Owner, Admin, Member, and Viewer, with permission checks on relevant actions.	Each role can perform permitted actions and is blocked from prohibited actions in test cases.
FR-WS-06	Must	The system shall allow authorized administrators to change a member's role or remove the member.	Membership changes are saved, take effect on subsequent protected requests, and are recorded in the activity log.
FR-WS-07	Should	The system shall display workspace/team membership and relevant member activity to authorized users.	An authorized user can see the member list and permitted activity without exposing private records.
FR-WS-08	Should	The system shall allow workspace owners/admins to configure workspace-level access and settings.	A saved setting affects only its intended workspace and cannot be changed by unauthorized members.

5.4 Projects, Templates, and Project Settings

ID	Priority	Functional requirement	Verification / acceptance condition
FR-PROJ-01	Must	The system shall allow authorized users to create projects inside a personal space or workspace as supported by the data model.	A project is created with required details and is visible only to users with access.
FR-PROJ-02	Must	The system shall allow authorized users to view and update project details and settings.	Updates persist and the access rules for the project remain enforced.
FR-PROJ-03	Must	The system shall allow a project owner/manager to configure board columns and their order.	The board displays the configured columns in the saved order.
FR-PROJ-04	Should	The system shall provide reusable starter templates, including Basic, Agile, and Scrum where configured.	Selecting a template creates a project with the template's predefined structure without overwriting unrelated projects.
FR-PROJ-05	Should	The system shall allow authorized users to save or reuse project/task templates.	A reusable template can be selected to create new work with the template's saved structure.
FR-PROJ-06	Must	The system shall allow an authorized user to archive a project and restore it where permitted.	Archived projects are distinguished from active projects; authorized restoration returns the project to active use.
FR-PROJ-07	Must	The system shall enforce project-level membership and access settings.	Users without project access cannot view or modify project records through direct URLs or server requests.
FR-PROJ-08	Should	The system shall support project milestones and an overall project progress indicator.	Milestones can be created/viewed and project progress is displayed using a documented calculation.

5.5 Tasks, Status Workflow, and Checklists

ID	Priority	Functional requirement	Verification / acceptance condition
FR-TASK-01	Must	The system shall allow an authorized user to create, view, edit, and delete or archive tasks according to permissions.	Valid task changes persist; unauthorized or invalid changes are rejected.
FR-TASK-02	Must	A task shall support a title and description, status, priority, and association with a project or personal workspace as applicable.	Saved tasks display their supported fields correctly.
FR-TASK-03	Must	A task shall support due date, time estimate, and optional start date where relevant to scheduling.	Valid dates and estimates are saved and displayed in task and planning views.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-TASK-04	Must	A task shall support priority values Low, Medium, High, and Critical.	A user can set a permitted priority and filter or sort tasks by it.
FR-TASK-05	Must	A task shall support one or more assignees where collaboration mode permits.	Authorized users can assign/unassign task members and assignment changes are recorded.
FR-TASK-06	Must	A task shall support colored labels/tags for categorization.	Authorized users can add/remove labels and use them in filters.
FR-TASK-07	Should	The system shall support task-specific custom fields configured for a project or workspace.	Configured custom fields can be entered, saved, displayed, and filtered where the field type supports filtering.
FR-TASK-08	Must	A task shall support subtasks or checklist items and show completion progress.	Users can complete/reopen checklist items and the displayed progress matches the saved items.
FR-TASK-09	Must	The system shall validate task status transitions against the configured workflow.	Valid transitions succeed; prohibited transitions are rejected or require the configured permission.
FR-TASK-10	Should	The system shall allow authorized users to copy or move a task between eligible projects.	The copied/moved task is linked to the destination and access permissions are rechecked.
FR-TASK-11	Should	The system shall support task archiving and restoration where permitted.	Archived tasks are removed from default active views and can be restored by an authorized user.
FR-TASK-12	Should	The system shall support recurring-work task templates or reuse of an existing task's structure.	A new task can be created from a template without duplicating unrelated comments or time records.
FR-TASK-13	Should	The system shall display task details, assignments, due dates, estimates, checklist progress, and activity in a consolidated task view.	A permitted user can inspect the task's main information from one details screen.

5.6 Kanban Boards and Workflow Visualization

ID	Priority	Functional requirement	Verification / acceptance condition
FR-KAN-01	Must	The system shall display project tasks on an interactive Kanban board grouped by status columns.	Tasks appear in the correct columns based on their persisted status.
FR-KAN-02	Must	The system shall allow users with permission to move a task between columns using drag-and-drop or an accessible alternative.	A move updates the task status and remains correct after a page refresh.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-KAN-03	Must	The system shall provide visual feedback when a task is moved or a move is rejected.	The user receives a clear success or validation result without a misleading final state.
FR-KAN-04	Should	The system shall support Work-in-Progress (WIP) limits for configured columns.	The system shows when a column limit is reached and applies the configured block/warning rule.
FR-KAN-05	Should	The system shall support swimlanes or an equivalent secondary grouping of tasks.	Users can group tasks by a configured category such as assignee or priority.
FR-KAN-06	Should	The system shall keep the board synchronized with updates made by other authorized users in near real time.	A second user sees a permitted task change without manually reloading, subject to connection availability.

5.7 Comments, Attachments, and Collaboration

ID	Priority	Functional requirement	Verification / acceptance condition
FR-COL-01	Must	Authorized users shall be able to add comments to a task.	A comment is saved with its author and timestamp and is shown to authorized project members.
FR-COL-02	Should	The comment interface shall support basic rich-text formatting where enabled.	Supported formatting is rendered safely and does not execute submitted scripts.
FR-COL-03	Should	Users shall be able to mention workspace/team members in a comment.	A valid mention is rendered and generates the configured notification to the mentioned user.
FR-COL-04	Must	Authorized users shall be able to upload supported task attachments.	Allowed files are stored and linked to the task; invalid or disallowed uploads are rejected with a clear message.
FR-COL-05	Must	The system shall apply access checks when a user views or downloads an attachment.	A user without access to the parent record cannot retrieve the file by guessing or reusing its URL.
FR-COL-06	Should	The system shall display active-viewing indicators for collaborators where real-time presence is enabled.	Authorized users can see that another collaborator is viewing the relevant item; stale presence expires.
FR-COL-07	Should	The system shall handle concurrent edits without silently overwriting a newer saved version.	A conflicting update is detected and the user is prompted to refresh, compare, or retry as supported.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-COL-08	Must	The system shall update supported collaboration views in real time through the configured live-communication mechanism.	A permitted change is broadcast to relevant connected users and is not sent to unauthorized users.

5.8 Task Handover Notes — Agreed Extension

ID	Priority	Functional requirement	Verification / acceptance condition
FR-HAND-01	Should	An authorized user shall be able to create a handover note when a task is transferred or reassigned.	The note records the current progress, blockers, and next steps, and is linked to the task.
FR-HAND-02	Should	The system shall display handover history to users who can access the task.	Authorized users can view prior handover notes with author and timestamp.
FR-HAND-03	Should	A handover workflow shall preserve the task's existing comments, attachments, and activity history.	Reassignment does not remove or detach the existing task records.

5.9 Calendar, Gantt, Dependencies, and Planning

ID	Priority	Functional requirement	Verification / acceptance condition
FR-PLAN-01	Must	The system shall provide a calendar view of authorized tasks based on their relevant dates.	Tasks with supported dates appear on the correct dates and can be opened from the calendar.
FR-PLAN-02	Must	The system shall provide a Gantt/timeline view for tasks in a project.	Tasks are displayed with their scheduled start/end dates or supported duration representation.
FR-PLAN-03	Should	Authorized users shall be able to create and remove task dependencies, including finish-to-start dependencies.	A dependency is saved and displayed between the correct tasks.
FR-PLAN-04	Should	Authorized users shall be able to adjust task dates or duration from the timeline.	A permitted timeline change persists to the task and is reflected in other planning views.
FR-PLAN-05	Should	The system shall visually identify dependency conflicts or scheduling constraints.	A conflict is shown when a configured rule is violated; the system does not silently change unrelated tasks.
FR-PLAN-06	Could	The Gantt view shall identify the critical path where the necessary scheduling data and algorithm are available.	A test project with known dependencies displays the expected critical path.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-PLAN-07	Should	The system shall support project milestones in planning views.	A milestone is visible in the relevant timeline/calendar representation.

5.10 Time Tracking and Timesheets

ID	Priority	Functional requirement	Verification / acceptance condition
FR-TIME-01	Must	A permitted user shall be able to start, pause, resume, and stop a timer against a task.	Timer state transitions are valid and the resulting elapsed time is persisted on the server.
FR-TIME-02	Must	The system shall prevent inconsistent active timer sessions for the same user according to the configured timer rule.	Attempts to start a conflicting timer are handled with an explicit message or the configured safe behavior.
FR-TIME-03	Must	A permitted user shall be able to add a manual time entry to an eligible task.	A valid entry records the duration, task, user, and date; invalid durations are rejected.
FR-TIME-04	Should	The system shall allow permitted users to edit or delete time entries subject to the configured audit policy.	Changes are authorized, validated, and recorded where required.
FR-TIME-05	Must	The system shall show time totals for tasks and support estimated-versus-actual comparisons.	Totals match the saved entries and are displayed with the task estimate when present.
FR-TIME-06	Should	The system shall provide timesheet views that summarize time by user, task, project, and date range as permitted.	A selected date range returns the expected entries and totals for the current user's access scope.
FR-TIME-07	Should	The system shall export timesheet data in at least one agreed downloadable format.	An export contains the selected authorized records and totals in a usable file.
FR-TIME-08	Should	The system shall provide project/team time reports to users with reporting permission.	Authorized users can compare tracked time across eligible tasks, members, and projects.

5.11 Dashboard Analytics and Rule-Based Project Health

ID	Priority	Functional requirement	Verification / acceptance condition
FR-DASH-01	Must	The system shall show task counts grouped by status.	The displayed values match the eligible task records used by the report.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-DASH-02	Must	The system shall show project progress using an explicit, documented calculation.	The displayed progress is reproducible from the underlying task/status rules.
FR-DASH-03	Should	The system shall show workload summaries by team member to authorized users.	The report counts eligible assignments within the selected workspace/project and date scope.
FR-DASH-04	Must	The system shall identify overdue tasks in task lists and relevant dashboards.	A task is highlighted when its due date has passed and it is not in a configured completed state.
FR-DASH-05	Could	The system shall provide sprint velocity and burndown/burnup charts when sprint data is configured.	A sample sprint with known data generates matching chart values.
FR-DASH-06	Should	The system shall provide rule-based project health alerts based on explicit conditions, such as overdue work, blocked dependencies, or tasks with no recent update.	When a configured condition is met, the system creates or displays the defined alert; no AI inference is used.
FR-DASH-07	Should	Authorized users shall be able to review, acknowledge, or resolve a project health alert.	The alert status and response are saved and visible to authorized project users.

5.12 Notifications and Reminders

ID	Priority	Functional requirement	Verification / acceptance condition
FR-NOT-01	Must	The system shall provide an in-app notification center.	A user can view notifications relevant to their account and access the associated authorized record.
FR-NOT-02	Must	The system shall notify relevant users about task assignment and important task changes.	A configured event creates a notification for the intended recipients only.
FR-NOT-03	Should	The system shall provide due-date reminders based on configured timing rules.	Eligible tasks trigger a reminder at the configured time and completed tasks follow the agreed suppression rule.
FR-NOT-04	Should	The system shall support email notifications where email delivery is configured.	Eligible events generate an email to the intended recipient, with delivery failures handled without corrupting the source action.
FR-NOT-05	Should	Users shall be able to configure supported notification preferences.	Preference changes persist and affect future notifications according to the preference rules.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-NOT-06	Could	The system shall support daily or weekly digest summaries.	A digest summarizes eligible updates for the selected period and respects recipient preferences.
FR-NOT-07	Must	Users shall be able to mark notifications as read or unread.	The notification state is persisted and reflected across the user's supported views.

5.13 Search, Filters, Saved Views, and Shortcuts

ID	Priority	Functional requirement	Verification / acceptance condition
FR-SEARCH-01	Must	The system shall provide global search across authorized projects and tasks.	Matching authorized results are returned; inaccessible records are not revealed in results or counts.
FR-SEARCH-02	Must	Users shall be able to filter tasks by status, priority, assignee, date, and labels where those fields exist.	Applying and clearing filters returns the expected result set.
FR-SEARCH-03	Should	Users shall be able to combine supported filters and sort the results.	Combined conditions use the documented logic and results remain within the user's access scope.
FR-SEARCH-04	Should	Users shall be able to save, rename, reuse, and delete personal or permitted shared views.	A saved view restores its filters/sort settings; shared views follow their defined visibility rules.
FR-SEARCH-05	Could	The application shall provide keyboard shortcuts for selected frequent actions and a reference page or overlay.	Supported shortcuts perform the stated actions and do not intercept typing in unrelated input fields.

5.14 Rule-Based Automation and Webhooks

ID	Priority	Functional requirement	Verification / acceptance condition
FR-AUTO-01	Should	An authorized user shall be able to configure automation rules using available triggers, conditions, and actions.	A saved rule is validated and its enabled/disabled status is shown clearly.
FR-AUTO-02	Should	The system shall support an automation that completes or moves a task when all required checklist items are completed, if configured.	Meeting the configured condition executes the configured action once according to the rule's duplicate-execution policy.
FR-AUTO-03	Should	The system shall support configured notifications when a due date approaches.	The rule triggers according to the configured lead time and recipient policy.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-AUTO-04	Could	The system shall support rule-based auto-assignment using configured labels or columns.	Only matching tasks are assigned and the selected rule's audit history is recorded.
FR-AUTO-05	Could	The system shall support configured webhook integrations for selected events.	A configured event sends the documented payload to the configured destination, and failure is logged for review.
FR-AUTO-06	Must	The system shall validate automation permissions and prevent unauthorized rules from modifying records.	A user cannot configure or trigger actions beyond their workspace/project permissions.

5.15 Activity Log and Audit Trail

ID	Priority	Functional requirement	Verification / acceptance condition
FR-AUD-01	Must	The system shall record important actions, including relevant creates, updates, status changes, assignments, membership changes, and archival actions.	A performed action creates an audit record containing the actor, action, affected record, and timestamp when applicable.
FR-AUD-02	Must	The system shall allow authorized users to view activity history for a project/task.	The displayed history is limited to records the user is allowed to access.
FR-AUD-03	Should	Authorized users shall be able to filter activity by user, task/project, date, or action type where supported.	Filters return the expected audit records and can be cleared.
FR-AUD-04	Should	Authorized users shall be able to export permitted activity records.	The export contains only records available to the requesting user and includes relevant timestamps and action details.
FR-AUD-05	Must	The system shall prevent ordinary users from silently altering or deleting audit records through normal product functions.	Standard user operations cannot overwrite the history of a previously recorded event.

5.16 Clients, Client Portal, and Project Live Room — Agreed Extensions

ID	Priority	Functional requirement	Verification / acceptance condition
FR-CLIENT-01	Should	An authorized freelancer/project owner shall be able to create or manage a client record and link it to selected projects.	The client is linked only to the intended projects and is not granted workspace-wide access by default.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-CLIENT-02	Should	The system shall allow authorized users to invite a client to a client-facing portal for a selected project.	The invitation grants access only after the configured acceptance/sign-in process and only to explicitly shared information.
FR-CLIENT-03	Must	The client portal shall show the client's authorized project progress and selected deliverables or milestones.	A client cannot view internal-only tasks, comments, files, time entries, or other unshared data.
FR-CLIENT-04	Should	The system shall allow authorized project users to control which project information is shared externally.	Changes to sharing permissions take effect on subsequent portal requests and are audited.
FR-CLIENT-05	Should	The Project Live Room shall collect approved progress updates, client-safe discussions, shared deliverables, and relevant project status in one view.	A user sees only the live-room content enabled for that user and project.
FR-CLIENT-06	Should	The system shall distinguish internal comments from client-visible messages.	Internal-only content is not exposed in the client portal or client notifications.

5.17 Deliverable Review and Change Requests — Agreed Extensions

ID	Priority	Functional requirement	Verification / acceptance condition
FR-DEL-01	Should	Authorized team members shall be able to create a deliverable record and associate it with a project or task.	The deliverable stores its name, description, owner, status, and related file/link as supported.
FR-DEL-02	Should	The system shall retain versions of a deliverable when updated.	A new version does not silently replace the previous version's record and authorized users can identify the current version.
FR-DEL-03	Should	An authorized reviewer/client shall be able to approve a deliverable or request changes when permitted.	The decision, reviewer, timestamp, and optional feedback are recorded.
FR-DEL-04	Should	The system shall display deliverable review status and feedback to authorized participants.	The current status and latest review feedback are visible to permitted users only.
FR-CHG-01	Should	An authorized user shall be able to submit a project change request describing the requested change and its reason.	The request is saved with a unique identifier, requester, date, and initial status.
FR-CHG-02	Should	The system shall allow authorized reviewers to assess the potential scope, schedule, and cost impact before approving a change request.	Impact fields can be recorded and reviewed before an approval decision.

ID	Priority	Functional requirement	Verification / acceptance condition
FR-CHG-03	Should	The system shall allow a change request to be approved, rejected, or returned for clarification by authorized reviewers.	The decision, reviewer, timestamp, and comments are recorded, and unauthorized users cannot decide.
FR-CHG-04	Should	An approved change request shall not silently modify project data; any resulting schedule/scope/task updates must be confirmed and auditable.	The applied changes match the approved request and related audit events are present.

5.18 Platform Administration, Usage, and Data Management

ID	Priority	Functional requirement	Verification / acceptance condition
FR-ADMIN-01	Must	The platform shall provide an administration area accessible only to platform administrators.	Non-admin accounts cannot access the area or invoke its protected operations.
FR-ADMIN-02	Should	Platform administrators shall be able to view and manage user accounts according to platform policy.	Authorized account-management changes are validated and recorded.
FR-ADMIN-03	Should	Platform administrators shall be able to view usage statistics and high-level platform activity.	Statistics reflect the documented data scope and do not expose unnecessary private content.
FR-ADMIN-04	Should	Authorized administrators shall be able to configure supported system settings.	Changes are validated, saved, and limited to authorized administrators.
FR-ADMIN-05	Could	The platform shall provide subscription/billing management if subscriptions are included in the approved release scope.	The configured subscription state and associated account are displayed accurately; unsupported payment handling is not simulated as complete.
FR-ADMIN-06	Should	Authorized users shall be able to export their permitted data.	The export contains selected authorized records and excludes inaccessible data.
FR-ADMIN-07	Should	The system shall support backup/restore procedures for platform data as part of operational administration.	A documented backup can be verified and a restore can be tested in a controlled environment.

5.19 Interface and Accessibility Functions

ID	Priority	Functional requirement	Verification / acceptance condition
FR-UI-01	Must	The application shall provide a responsive interface for supported desktop, tablet, and mobile screen widths.	Core user flows remain usable without unintended horizontal overflow at agreed test sizes.
FR-UI-02	Should	The application shall provide dark and light visual modes.	The selected mode is applied consistently to supported pages and controls.
FR-UI-03	Must	Core functions shall be usable with keyboard navigation where technically applicable.	Interactive controls are reachable in a logical order and display a visible focus indicator.
FR-UI-04	Should	The interface shall follow the team's agreed accessibility target, using WCAG 2.1 as a reference.	The team tests applicable keyboard, form-label, contrast, and status-feedback criteria and documents known gaps.
FR-UI-05	Should	The system shall provide clear validation, success, warning, and error messages for user actions.	Failed operations explain what needs attention without exposing technical secrets.
FR-UI-06	Could	The system shall provide customizable dashboard widgets and keyboard shortcut references for power users.	Available customizations and shortcut help operate as documented.

6. Core Business Rules

1. **Personal-first access:** A registered user can use personal workspace features without creating or joining a team.
2. **Least privilege:** A user can view or modify only records permitted by the account, workspace, project, client-sharing, and role permissions applicable to the request.
3. **Client isolation:** Client accounts only see content explicitly shared for their project; internal-only tasks, discussions, files, and time data remain hidden.
4. **Task status integrity:** Status changes must follow the configured workflow and must be validated on the server, not only in the browser.
5. **Task completion:** The project must define which statuses count as completed for progress, overdue highlighting, and reports; the same definition should be used consistently.
6. **Time accuracy:** Timer sessions and manual entries must be persisted and validated server-side. Displayed totals must derive from saved time records.
7. **Auditability:** Important changes, permission changes, review decisions, change-request decisions, and automation actions should leave appropriate history.
8. **Safe automation:** Rules execute only for authorized scopes and configured conditions. Repeated event processing must not accidentally duplicate the same action.
9. **Change control:** A change request is not considered approved until an authorized reviewer records a decision. Downstream project updates must be traceable to the approved decision.
10. **Deliverable review:** Approval or a request for changes must identify the reviewer and the version being reviewed.

11. **Real-time access control:** SignalR/live updates must follow the same authorization scope as ordinary page requests.
12. **No AI behavior:** Search, health alerts, reports, reminders, and automations operate from explicit data, filters, and rules. They do not depend on AI models.

7. Required Inputs and Outputs by Main Workflow

Workflow	Main inputs	Expected outputs
Registration/sign-in	Account details and credentials	Authenticated account/session or validation feedback
Create project	Project name, owner/space, optional template and settings	Project record, initial structure, and access assignments
Create/update task	Title, description, status, priority, due date, assignees, labels, estimate	Persisted task, updated views, audit entry, and relevant notifications
Move task on board	Task, target column, ordering information	Updated status/order or a validation message
Track time	Task, timer action or manual duration, date	Persisted time entry/session and updated totals
Invite member/client	Invitee contact, target workspace/project, role and sharing scope	Invitation status and permission assignment after acceptance
Review deliverable	Deliverable version, decision, reviewer feedback	Recorded approval/request-for-changes decision and updated status
Change request	Description, reason, scope/schedule/cost impact, decision	Request record and auditable decision; downstream changes only after approval
Respond to health alert	Alert, acknowledgement/resolution status, optional note	Updated alert state and activity entry
Generate report/export	Scope, filters, date range, requested format	Authorized report data or downloadable export

8. Acceptance and Verification Approach

The team should verify functional requirements through a mix of:

- **Positive tests:** valid users complete the intended action and see the expected saved result.
- **Negative/permission tests:** invalid inputs and unauthorized requests are rejected without leaking data.
- **Persistence tests:** saved state survives navigation and page refreshes.
- **Cross-module tests:** a task change is reflected in relevant board, calendar/Gantt, search, notifications, reports, and activity history where applicable.
- **Real-time tests:** eligible connected users receive updates, while unauthorized users do not.
- **Export tests:** exported data matches the selected filters and the requesting user's access.
- **Rule tests:** automation and health alerts trigger only when their explicit conditions are met.

Each requirement should be assigned an implementation status and linked to one or more test cases during implementation. The requirements themselves do not imply that the tests have already passed.

9. Team Ownership and Traceability

The following is the current proposed module ownership. Cross-module requirements require coordination and integration testing.

Member	Primary ownership	Requirement areas
Mostafa Mohamed Mahmoud Mohamed	Team lead, authentication/authorization foundation, personal productivity and navigation	FR-AUTH, FR-PER, FR-SEARCH (global search and saved views), integration and release coordination
Ahmed Reda Ahmed Sedik	Workspaces, teams, memberships, roles, invitations, projects and project settings/templates	FR-WS, FR-PROJ, workspace/project access integration
Alaa Essam Hamed Abdelkader	Kanban, task business logic, task details/lifecycle, subtasks, comments, attachments, task activity and handover notes	FR-TASK, FR-KAN, FR-COL, FR-HAND, task-level business rules
Ahd Ammar Eisa El-Baqary	Calendar/Gantt, dependencies, time tracking/timesheets, reporting and rule-based health alerts	FR-PLAN, FR-TIME, FR-DASH and time/report integration
Eman Mohamed Masad El-Metwally	Client management and sharing, client portal/live room, deliverables/reviews and change requests	FR-CLIENT, FR-DEL, FR-CHG
Shared / coordinated	Notifications, audit log, automation, platform administration, UI consistency, security, testing and deployment	FR-NOT, FR-AUTO, FR-AUD, FR-ADMIN, FR-UI and cross-cutting acceptance tests

The public marketing pages are handled separately and are not part of the module ownership listed above.

10. Open Decisions for Analysis & Design

The following decisions need confirmation before implementation details are frozen:

1. Whether personal projects use the same project entity and board model as workspace projects or use a personal-space relationship.
2. Which file types and maximum file size will be allowed for attachments and deliverables.
3. Whether comments, task updates, and deliverable reviews support editing/deletion, and the retention/audit rules that apply.
4. The exact progress calculation and the statuses that count as complete.
5. Whether a user may run only one active timer globally or one active timer per project/task context.
6. Which reports and export formats are mandatory for the first demonstrable release.
7. Whether subscription/billing management is included in the delivered scope or documented as a future/optional platform capability.
8. Which automation rules and webhook event types can realistically be implemented within the project schedule.
9. Whether clients need their own account registration or can only join through an invitation.
10. Which notifications are in-app only versus email, based on the configured email provider and delivery setup.
11. Whether critical-path calculation is feasible in the planned Gantt implementation timeframe.
12. The specific accessibility checks and supported browser/screen-size matrix used for acceptance.

11. Sign-off

Role	Name	Review status	Date / comments
Team Lead	Mostafa Mohamed Mahmoud Mohamed	Pending review	
Workspace/Project Module	Ahmed Reda Ahmed Sedik	Pending review	
Task/Kanban Module	Alaa Essam Hamed Abdelkader	Pending review	
Planning/Reporting Module	Ahd Ammar Eisa El-Baqary	Pending review	
Client/Deliverables Module	Eman Mohamed Masad El-Metwally	Pending review	
Lecturer / Supervisor	To be completed	Pending feedback	

Document note: This is a proposed functional specification based on the TaskFlow project brief and the team's agreed product direction. Final scope and priorities should be reviewed before implementation. No AI functionality is included.